1.

Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

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Common Network Threats and Real-World Examples

Threat	Description	Real-World Example in an App/Service
Phishing	Deceptive attempts to steal sensitive data by masquerading as a trusted entity.	Receiving a fake SMS or WhatsApp message claiming your Paytm account is blocked and asking you to click a link to update KYC credentials.
DDoS (Distributed Denial of Service)	Overwhelming a server with traffic from multiple sources to make a service unavailable.	Flooding Instagram authentication servers with bot requests during a high-profile live event, causing temporary service outages for legitimate users.
Man-in-the-Middle (MitM)	Intercepting communication between two parties to eavesdrop or steal data.	A malicious actor setting up a rogue public Wi-Fi hotspot to intercept unencrypted traffic or session tokens when users access apps like WhatsApp on unsecured networks.

Threat	Description	Real-World Example in an App/Service
Malware	Malicious software designed to damage, disrupt, or gain unauthorized access to systems.	Downloading a fake modded APK file of Instagram (e.g., "Instagram Pro") from an untrusted third-party website that embeds keylogger spyware.
Password Attacks	Attempts to crack or steal user credentials using brute-force or credential stuffing.	Attackers using credential stuffing tools with leaked password databases from other breaches to automatically attempt logins on Paytm user accounts.

3.

List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

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A server room hosting high-traffic transactional applications like **Zomato** or **BookMyShow** faces several critical physical threats:

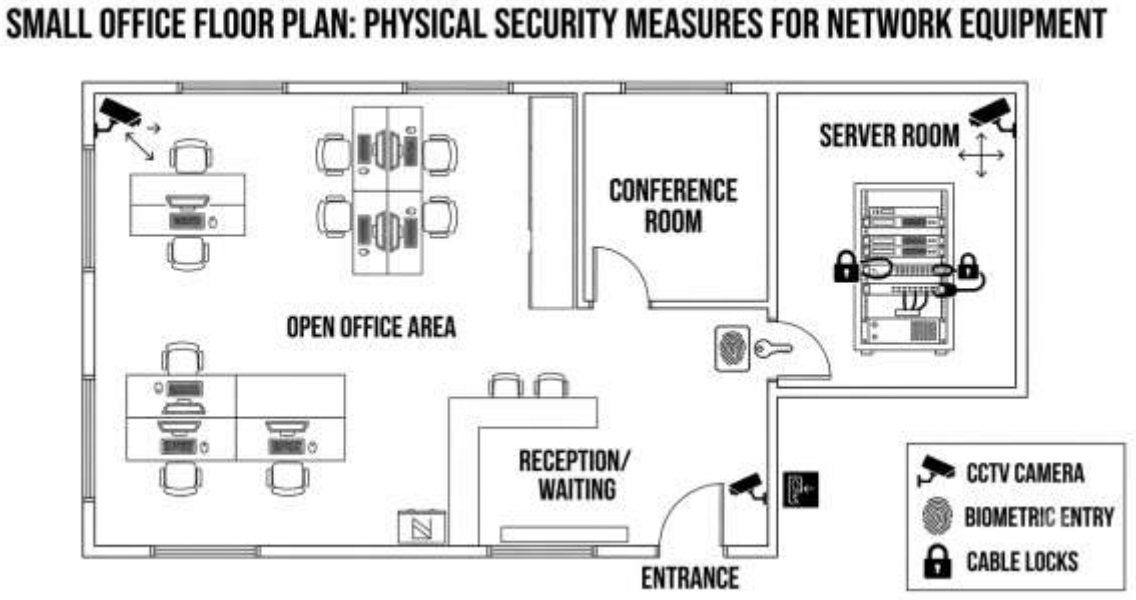
- **1. Unauthorized Physical Access (Tailgating or Intrusion)**

    - *Risk:* Unauthorized individuals entering the server room to tamper with hardware, steal physical storage drives, or connect rogue network devices.
    - *Mitigation:* Implement multi-factor biometric access control combined withmantrap doors and 24/7 CCTV surveillance.
- **2. Fire Hazards and Electrical Faults**
    - *Risk:* Overheating servers, short circuits, or overloaded power distribution units (PDUs) causing catastrophic fires that destroy critical infrastructure.
    - *Mitigation:* Install an early-warning aspiration smoke detection system (VESDA) paired with a clean-agent gaseous fire suppression system (like FM-200 or Novec 1230) that won't damage electronics.
- **3. Environmental Failures (HVAC and Water Leaks)**
    - *Risk:* Air conditioning failure leading to server overheating and shutdowns, or water leaks from overhead plumbing damaging server racks.
    - *Mitigation:* Deploy redundant precision cooling units and automated water-leak detection sensors installed beneath raised flooring.
- **4. Power Surges and Blackouts**
    - *Risk:* Sudden power loss interrupting live food delivery orders or movie ticket bookings, corrupting database transactions.
    - *Mitigation:* Install enterprise-grade Uninterruptible Power Supply (UPS) battery banks backed by an automatic-start diesel generator.

4.

Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

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5.

Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

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Here is a customized 7-action security checklist specifically tailored to protect a college computer lab environment from both network threats and physical breaches:

1. **Implement Network Segmentation (VLANs):** Isolate student lab workstations onto a separate VLAN away from administrative systems, faculty networks, and university backend databases to contain potential breaches.

2. **Enforce Port Security & 802.1X Authentication:** Configure switch port security to limit allowed MAC addresses per port and use 802.1X authentication to prevent students from plugging unauthorized laptops into wall ports.
3. **Secure Physical Hardware & Peripherals:** Anchor desktop computers and monitors to lab desks using security cables, lock server/rack cabinets containing network switches, and install security cameras covering all entry points.
4. **Deploy Deep Freeze and EDR Software:** Install endpoint detection and response (EDR) antivirus alongside disk-protection software (such as Deep Freeze) on lab PCs to automatically wipe malware or unauthorized system modifications upon reboot.
5. **Configure Strict Content Filtering & Firewalls:** Set up boundary firewalls and DNS/web content filtering to block malicious domains, torrenting, peer-to-peer file sharing, and high-risk gambling/phishing websites.
6. **Restrict USB and BIOS/UEFI Access:** Password-protect BIOS settings to disable bootable USB devices and disable USB mass storage device drivers via Group Policy to prevent malware sideloading or data exfiltration.
7. **Automate Patch Management & Student Guest Accounts:** Restrict students to limited-privilege guest accounts and automate operating system and application security patching during scheduled off-hours maintenance windows.